electrons to roam free among the photons, batting them to and fro as they interact with one another.

But this freedom comes to an abrupt end when the temperature of the universe falls below 3,000 degrees Kelvin (about half the temperature of the Sun's surface), and all the free electrons combine with nuclei. The marriage leaves behind a ubiquitous bath of visible light, forever imprinting the sky with a record of where all the matter was in that moment, and completing the formation of particles and atoms in the primordial universe.

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For the first billion years, the universe continued to expand and cool as matter gravitated into the massive concentrations we call galaxies. Nearly a hundred billion of them formed, each containing hundreds of billions of stars that undergo thermonuclear fusion in their cores. Those stars with more than about ten times the mass of the Sun achieve sufficient pressure and temperature in their cores to manufacture dozens of elements heavier than hydrogen, including those that compose planets and whatever life may thrive upon them.

These elements would be stunningly useless were they to remain where they formed. But high-mass stars fortuitously explode, scattering their chemically enriched guts throughout the galaxy. After nine billion years of such enrichment, in an undistinguished part of the universe (the outskirts of the Virgo Supercluster) in an undistinguished galaxy (the Milky Way) in an undistinguished region (the Orion Arm), an undistinguished star (the Sun) was born.

The gas cloud from which the Sun formed contained a sufficient supply of heavy elements to coalesce and spawn a complex inventory of orbiting objects that includes several rocky and gaseous planets, hundreds of thousands of asteroids, and billions of comets. For the first several hundred million years, large quantities of leftover debris in wayward orbits would accrete onto larger bodies. This occurred in the form of high-speed, high-energy impacts, which rendered molten the surfaces of the rocky planets, preventing the formation of complex molecules.

As less and less accretable matter remained in the solar system, planet surfaces began to cool. The one we call Earth formed in a kind of Goldilocks zone around the Sun, where oceans remain largely in liquid form. Had Earth been much closer to the Sun, the oceans would have